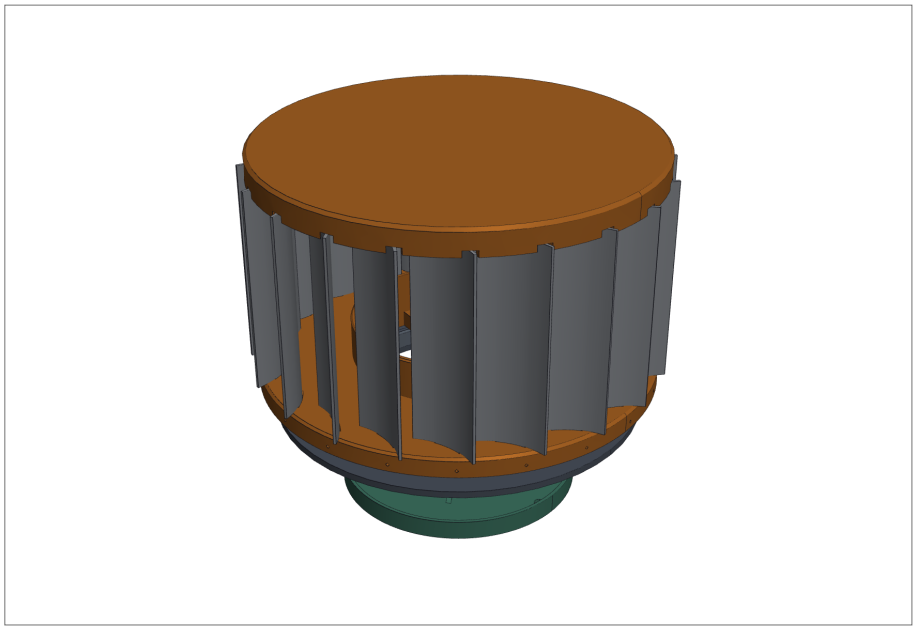
DOGHOUSE WHIRLYBIRD - CAD / FABRICATION PACKAGE

Rev DW-V2.0 - generated 2026-10-24 from the parametric build123d model (whirlybird/cad) and MASTER_PARAMETERS.json



COORDINATE SYSTEM / DATUMS

Z0 = finished roof surface at the rotor axis. +Z = rotor axis (roof to sky).
X/Y = plan. Blade 1 tab centre on +X axis; blades numbered CCW seen from above.
Datum A = shaft axis. Datum B = lower-hub top face (blade shoulder seat).
All STEP files share this origin; STL files are re-posed for printing.

Key result (from CAD)	Value
Installed height, roof to rotor top	169.7 mm (6.68 in) limit 190.5
Rotor OD (blade tips)	257.9 mm (10.15 in) target 266.7
Blades	20 cupped vanes, R45 x 5.4 mm deep
Hub OD upper / lower	248 / 248 mm
Cap separation (blade shoulders)	125.70 mm (= blade body length)
Blade chord angle from radial	55 deg, all same hand
Plan coverage / air channel	100.3 % (closed ring) / 11.5 mm
Rotor-to-stationary axial gap	6.0 mm (labyrinth 2.0 radial)
Bearing spacing / shaft length	50 mm / 198.8 mm
Throat (roof opening)	127 mm (5 in hole saw)
Height margin to 190.5 limit	20.8 mm

DESIGN GATES (18/18 PASS) - see sheet 12 and whirlybird/reports/design_checks.json

#	Gate (automated on the CAD)	Result
1	No unintended solid interference anywhere in the assembly	PASS
2	Rotating parts never touch stationary parts (min clearance >= 1.5 mm, excl. bearing internal	PASS
3	Lower hub face to stationary collar/support axial gap = 6 mm	PASS
4	Blade-to-blade air channel >= 10 mm (cupped blades, no collision)	PASS
5	Closed blade ring: plan coverage >= 100 % (no straight line of sight through the ring)	PASS
6	Blade-to-stationary clearance	PASS
7	Tab fits pocket with clearance on all sides and does not bottom out	PASS
8	Upper hub top face is one closed plane (no pocket breaks through)	PASS
9	Pocket walls to every exterior face >= 3.0 mm (top/bottom/OD)	PASS
10	Fixings reachable with the closed blade ring: lower set screw before blades, tie-rod nut fro	PASS
11	Assembly sequence collision-free (blades drop into lower hub, upper hub drops over tabs)	PASS
12	Blades positively trapped once hubs are fixed (axial + radial)	PASS
13	Installed height <= 190.5 mm	PASS
14	Assembly CAD top equals reported installed height	PASS
15	Rotor clears roof surface by >= 6 mm (at modelled pitch)	PASS
16	Every printed part fits the X1E bed in print orientation	PASS
17	Shaft length fits user's 209.55 mm tube (if bearings accept it)	PASS
18	All STL meshes watertight, manifold, single body, volume within 1 %, on bed	PASS

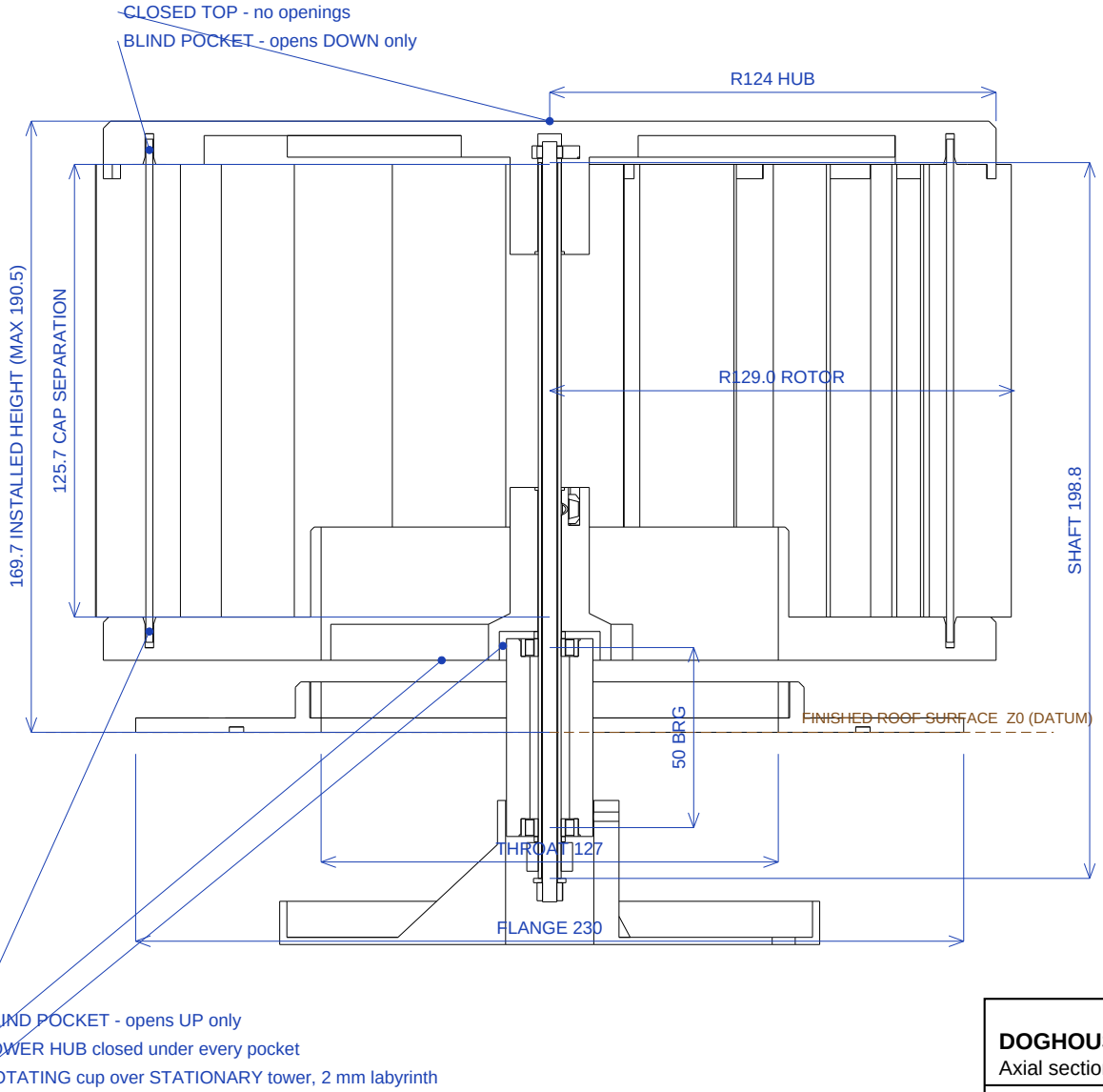
SHEET INDEX

1 Cover / datums / results
2 Axial section, height stack
3 Rotor vs stationary
4 Exploded assembly + order
5 True CAD views
6 Upper hub
7 Lower hub
8 Bearing support
9 Roof flash adapter
10 Blade 1:1 + forming
11 Coupons + tooling
12 Checks + calcs

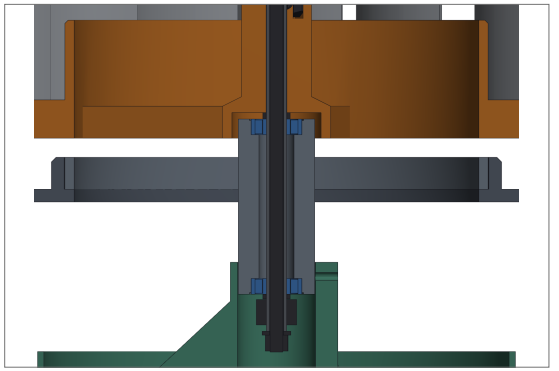
DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Cover, datums, results

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

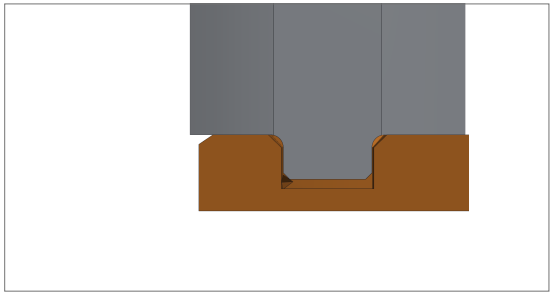
SHEET 2 - AXIAL HALF-SECTION (plane XZ, true CAD hidden-line projection, 1:2) + HEIGHT STACK



Bearing / labyrinth detail (render)



Lower pocket detail: tab, R2 roots, blind floor, weep



Height stack (Z, mm)	from	to	size
Roof to collar/support top	0	14.0	14.0
Rotor-to-stationary gap	14.0	20.0	6.0
Lower hub	20.0	32.0	12.0
Blade shoulders (cap sep.)	32.0	157.70	125.70
Upper hub	157.70	169.70	12.0
Bearing centres	-26.46	23.54	50.0
Shaft	-40.56	158.20	198.76

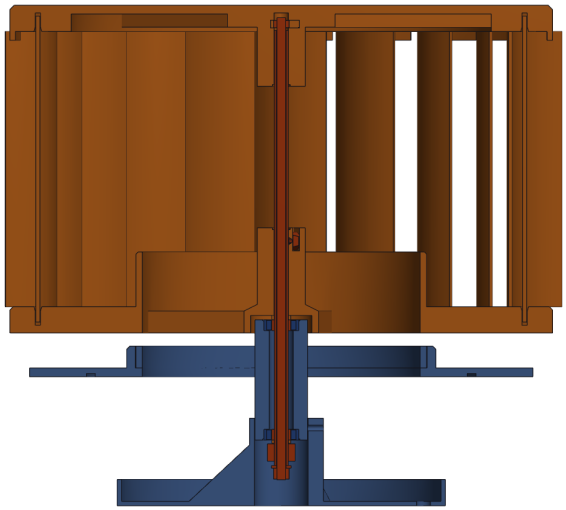
DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Axial section, height stack

REV DW-V2.0 UNITS mm SCALE 1:2 2026-10-24

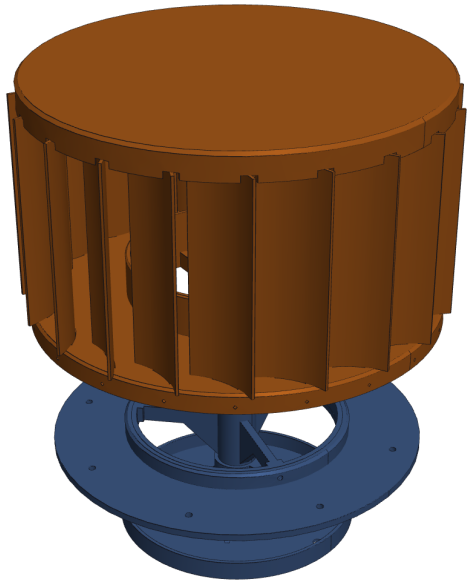
 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 3 - ROTOR vs STATIONARY: what turns, what is fixed to the roof, and how each attaches

SECTION - ORANGE rotating / BLUE stationary



Rotor (orange) lifted 70 mm off the stationary support (blue)



ROTATING part (orange)	Attaches to	By
Upper hub (closed top)	shaft + blades	M4 tie rod in tube -> captive nut; pulled onto shoulders
Lower hub (closed bottom)	shaft	1x M4 set screw in nut slot -> shaft dimple
20 blades	both hubs	tabs trapped in blind pockets (no fasteners)
Shaft 1/4 in TUBE + tie rod	bearing INNER races	nyloc nut + washer under the tube end
Top thrust washer	upper bearing inner race	loose on shaft, under lower-hub cup
Shaft collar + bottom washer	lower bearing inner race	collar set screw (uplift stop, 0.2 endplay)

STATIONARY part (blue)	Attaches to	By
Roof flash adapter	roof deck	8x #8 SS screws + sealant bead in groove
Bearing support (ring/spokes/tower)	adapter collar counterbore	rests on step; 3x M3 radial screws
2x R4-2RS bearing OUTER races	bearing support tower	pressed into seats (fit per coupon)
Rain tray (optional)	tower bottom (below roof)	slip-on standpipe + 1x M3 clamp screw

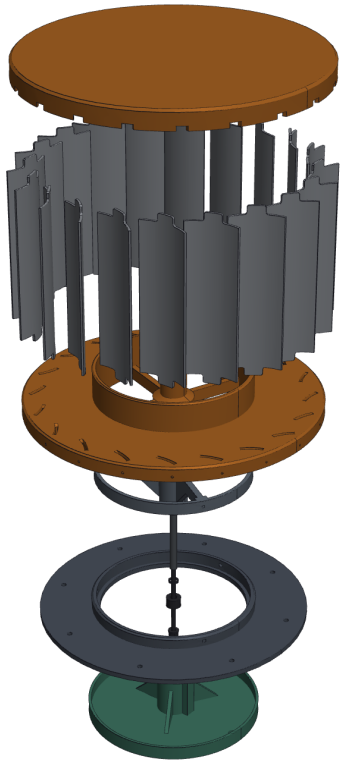
NO stationary part is fastened to either rotating hub. The only paths between rotor and stator are the two ball bearings. Minimum running clearances (from CAD): 6.0 mm lower hub to collar/support ring, 2.0 mm radial labyrinth (cup vs tower), 11.5 mm air channel blade to blade, >13 mm blade to any stationary part.
Service: remove 3 M3 screws -> the whole rotor + bearing cartridge lifts out of the roof adapter.

DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Rotor vs stationary

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

 Source: build123d parametric model + MASTER_PARAMETERS.json

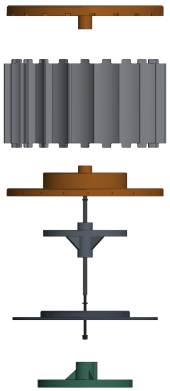
SHEET 4 - EXPLODED ASSEMBLY (true CAD) AND ASSEMBLY ORDER



ASSEMBLY ORDER (proved collision-free in CAD, gate 10):

1 Press both R4-2RS bearings into the support tower (top seat from above, bottom seat from below) - fit chosen from the fit coupon.
2 Bottom washer + 1/4 in shaft collar loosely on the shaft; slide shaft UP through both bearings from below.
3 Top thrust washer onto the shaft, on the upper inner race.
4 Lower hub down the shaft until its cup ceiling sits on the washer. Mark through the set-screw hole; spot-drill dimple (guide); M4 set screw + nut, threadlocker. Set collar for 0.2 mm endplay.
5 Drop each blade's lower tab straight DOWN into a lower-hub pocket (all 20 the same way: concave side as in the plan view).
6 Captive M4 nut into the upper hub side slot. Lower the hub straight DOWN over all upper tabs (lead-ins align them).
7 M4 rod up through the tube into the nut; washer + nyloc from below; snug until every shoulder seats. Blades trapped, no blade hardware.
8 Spin by hand: no rub, no rattle. Static-balance (tape on hub rim).
9 Fasten roof adapter (sealant + 8 screws); drop the rotor cartridge into the collar; 3x M3 radial screws. Optional tray underneath.

Exploded offsets are along +Z only, matching the real assembly direction: blades enter the lower hub from ABOVE; the upper hub comes DOWN over the upper tabs. No part has to be threaded over an already-bent blade.



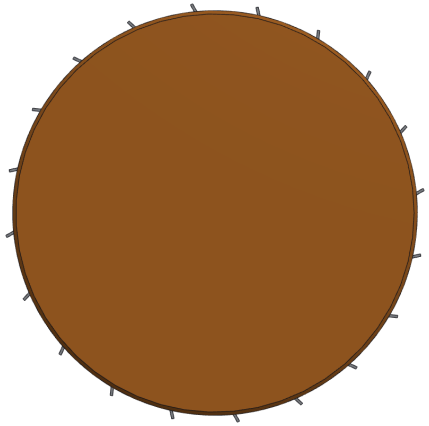
DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Exploded assembly

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

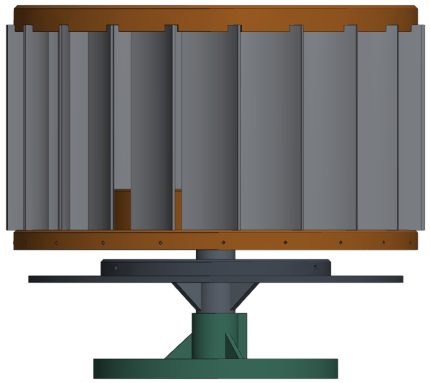
 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 5 - TRUE CAD-RENDERED VIEWS (tessellated build123d solids)

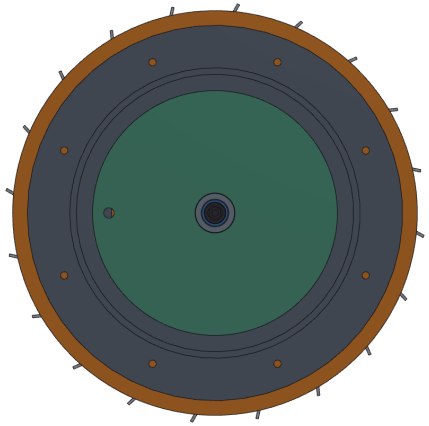
TOP



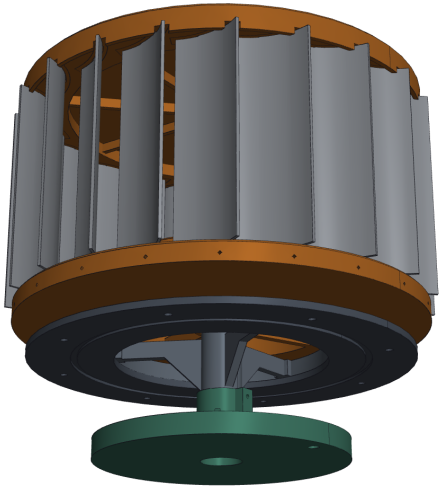
SIDE



UNDERSIDE



ISO FROM BELOW



DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR

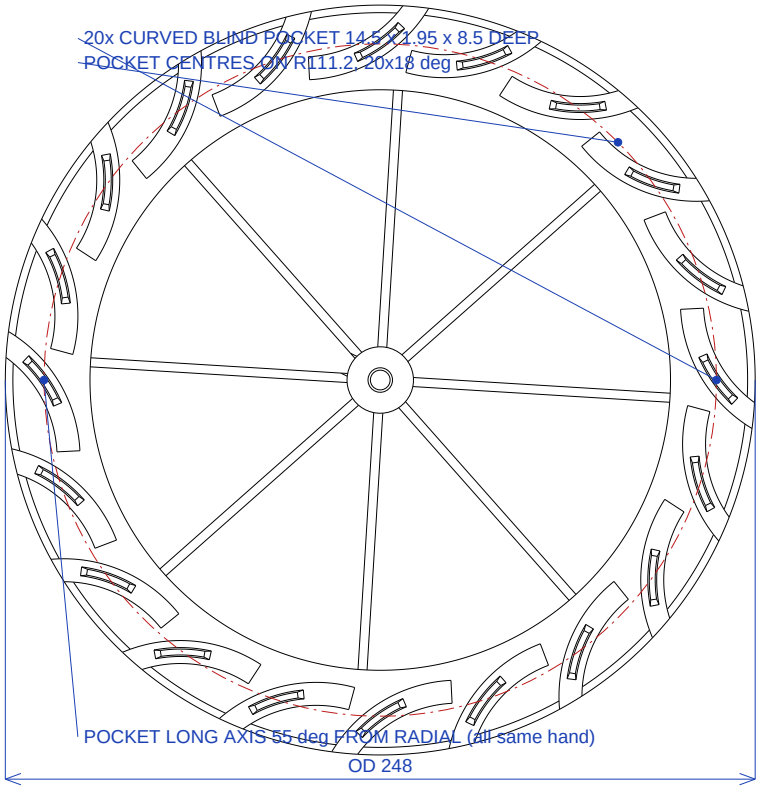
True CAD views

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

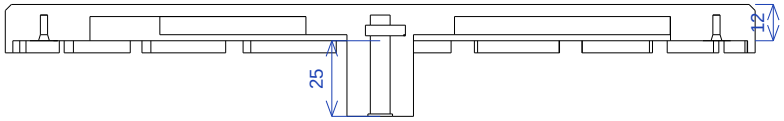
 Source: build123d parametric model + MASTER_PARAMETERS.json

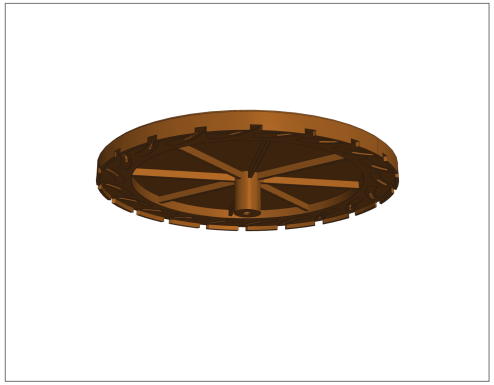
SHEET 6 - UPPER HUB CLOSED TOP (print TOP FACE DOWN)

PLAN FROM BELOW (pocket face) 1:2.5

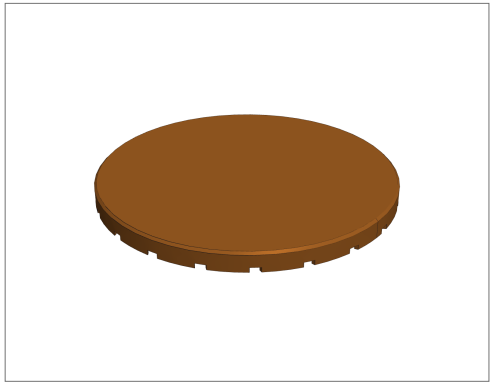


HALF SECTION XZ 1:2.5





pockets visibly BLIND



closed exterior TOP

Feature	Value
Pocket width (coupon-selected)	1.95 (try 1.85/1.95/2.05/2.15)
Pocket length / depth	14.5 / 8.5 (tab 14 x 7)
Mouth lead-in	ends 2.0 x45, sides 0.5
Intact wall over/under pocket	3.5 mm
Wall pocket->OD (mouth / deep)	5.60 / 7.38
Shaft bore	6.55 BLIND
Tie-rod nut	captive M4, side slot 7.3x3.5, Z 159.50
Pocket shape	curved R45 (follows the cupped tab)
Drip lip	2.5 x 4.0 notched at blades
Centre web / ribs	4.0 mm web, 8 ribs

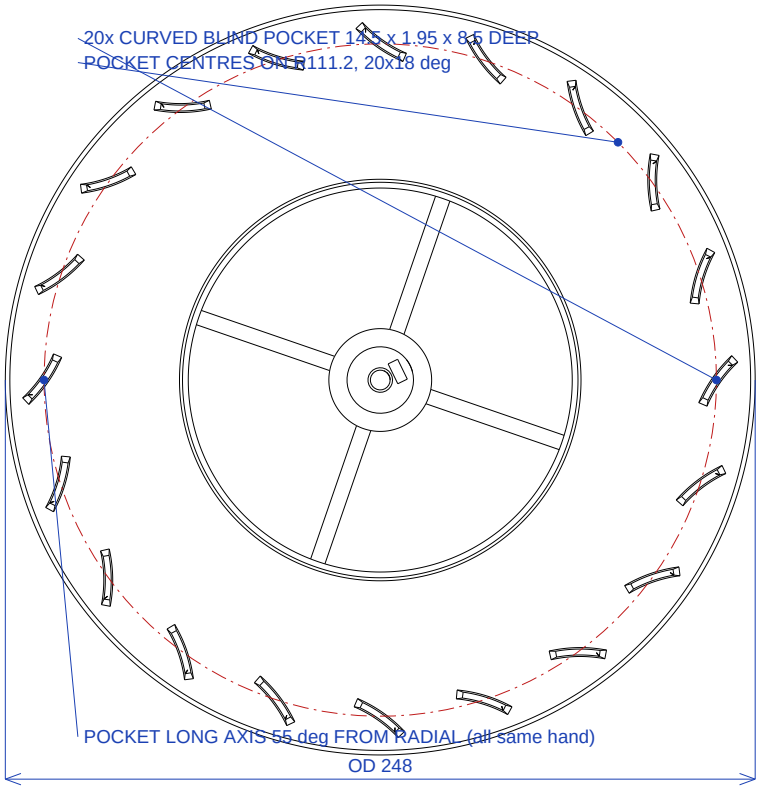
DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Upper hub

REV DW-V2.0 UNITS mm SCALE 1:2.5 2026-10-24

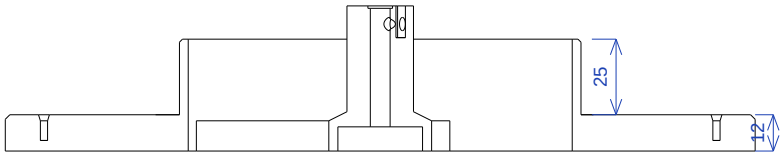
SHEET 6/12 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 7 - LOWER HUB CLOSED BOTTOM (print BOTTOM DOWN)

PLAN FROM ABOVE (pocket face) 1:2.5

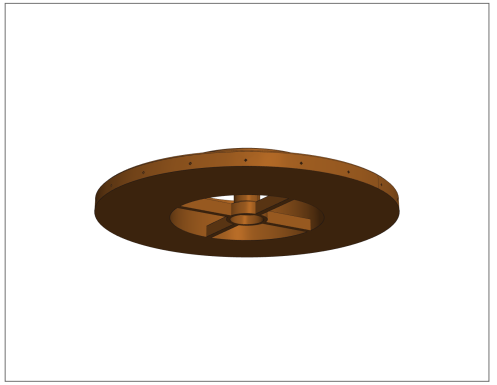


HALF SECTION XZ 1:2.5





pockets open UP only



closed exterior BOTTOM

Feature	Value
Pocket width (coupon-selected)	1.95 (try 1.85/1.95/2.05/2.15)
Pocket length / depth	14.5 / 8.5 (tab 14 x 7)
Mouth lead-in	ends 2.0 x45, sides 0.5
Intact wall over/under pocket	3.5 mm
Wall pocket->OD (mouth / deep)	5.60 / 7.38
Shaft bore	6.55 through boss
Set screw	M4 x6 cup, nut slot 7.3x3.5, Z 62.00
Pocket shape	curved R45 (follows the cupped tab)
Weep channels	8x 1.6 diamond, pocket floor -> OD, 2 deg fall
Chimney ID / spokes	127 / 4 x 5 mm
Labyrinth cup	ID 28, ceiling Z 28.0

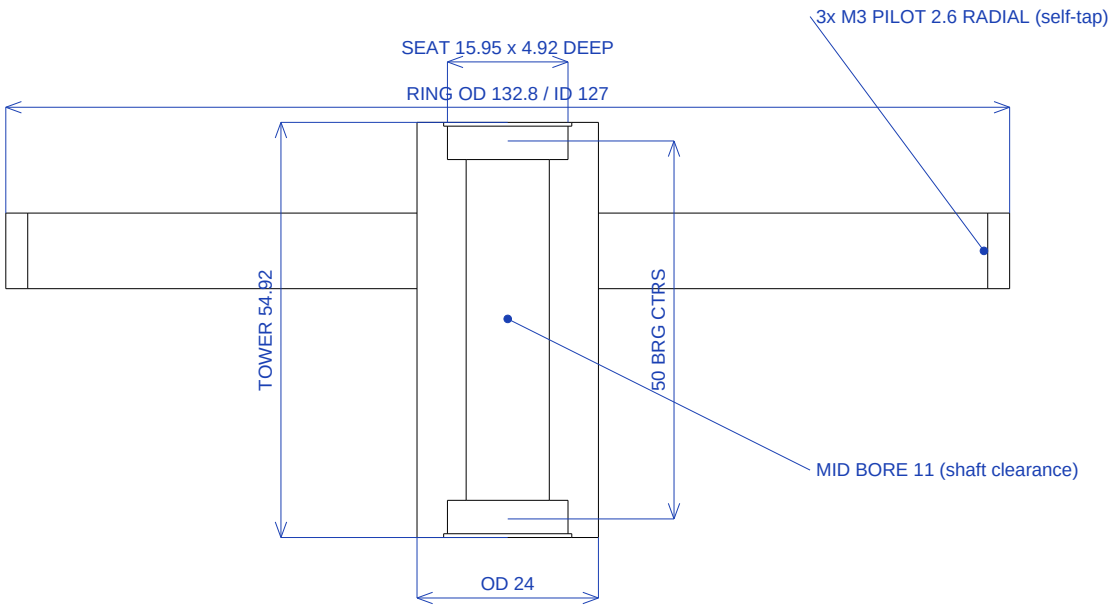
DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Lower hub

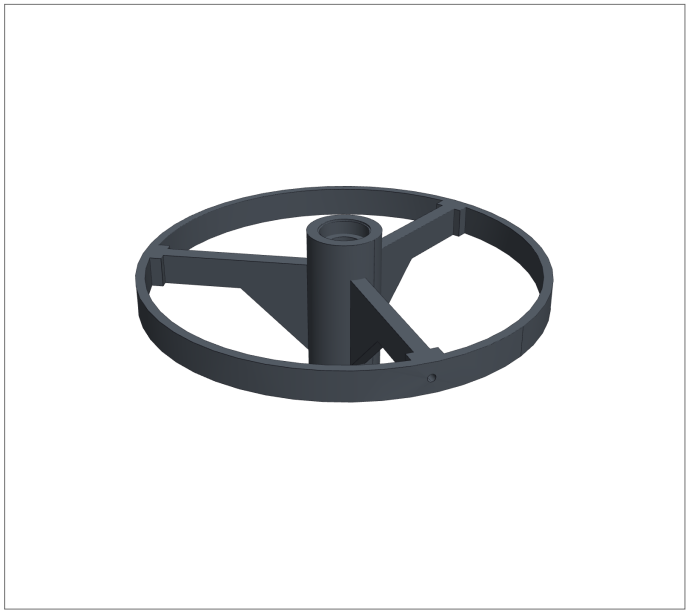
REV DW-V2.0 UNITS mm SCALE 1:2.5 2026-10-24

SHEET 7/12 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 8 - BEARING SUPPORT (STATIONARY) print TOWER-TOP DOWN, supports under ring/spokes

HALF SECTION XZ 1:1





Notes:
- Bearing OUTER races only are held here.
- Seat bores from fit coupon; do not print hard interference before testing.
- Gussets under each spoke stiffen the 43 mm tower below the ring.
- Ring top flush with collar top: 6 mm below the rotating lower hub.

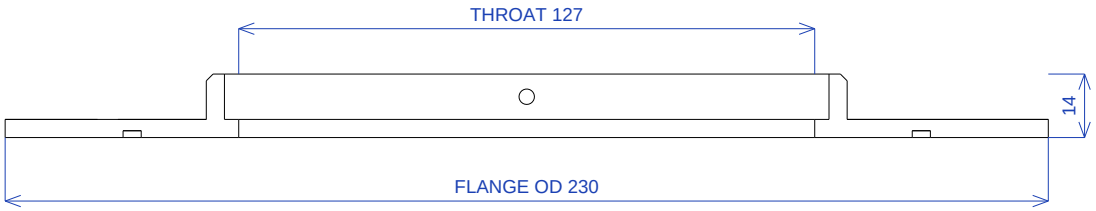
DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Bearing support

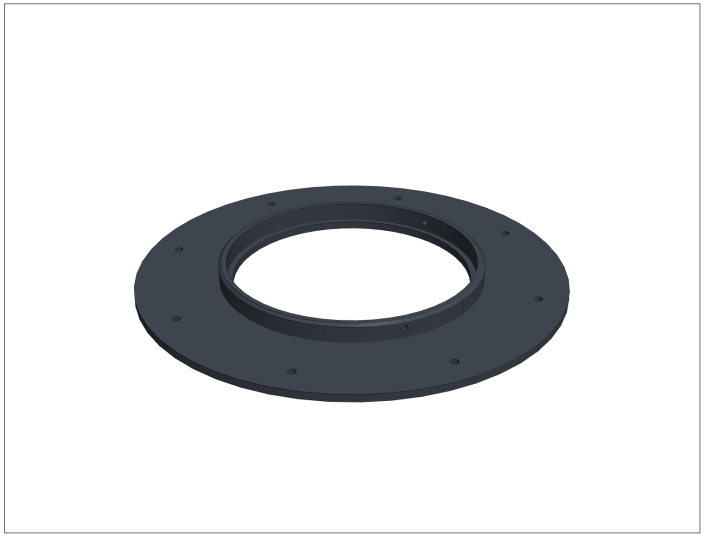
REV DW-V2.0 UNITS mm SCALE 1:1 2026-10-24

SHEET 8/12 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 9 - ROOF FLASH ADAPTER (STATIONARY) pitch modelled = 0.0 deg

HALF SECTION XZ 1:1.67





Roof pitch file	deg	Collar top Z	Installed height	<=190.5
roof_flash_adapter_pitch_0in12.stl	0.0	14.0	169.7	YES
roof_flash_adapter_pitch_2in12.stl	9.46	23.1	178.8	YES
roof_flash_adapter_pitch_3in12.stl	14.04	31.0	186.7	YES
roof_flash_adapter_pitch_4in12.stl	18.43	41.3	197.0	NO
roof_flash_adapter_pitch_5in12.stl	22.62	51.7	207.4	NO

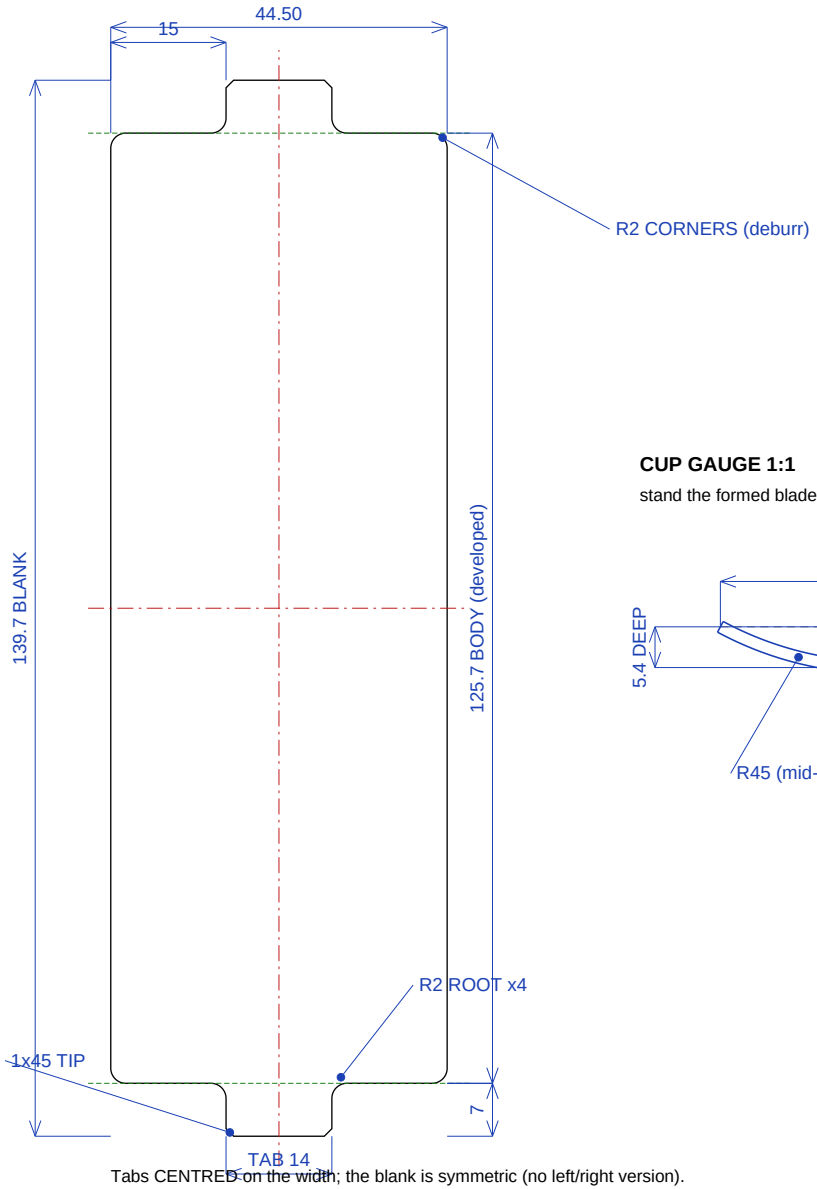
ROOF CUT-OUT: 127 mm (5 in) hole saw, centred. Seal the plywood cut edge with exterior paint.
FASTEN: 8x #8 x 3/4 in stainless pan-head wood screws with EPDM washers on a 200 mm circle.
SEAL: continuous bead of butyl / polyurethane in the 4 x 1.5 mm groove (R87),
plus a fillet bead round the flange edge on the UPHILL half. On shingles, slip the uphill half under the course above (no sealant needed there) - classic step-flash logic.
PITCH: real pitch unknown. Pitched files tilt the flange; the collar stays vertical. Steeper roofs lift the rotor to clear the uphill side; see the table for which pitches stay within 190.5 mm.
SUPPORT: bearing-support ring drops into the counterbore and rests on the step; 3x M3 screws radial.

DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Roof flash adapter

REV DW-V2.0 UNITS mm SCALE 1:1.67 2026-10-24

  Source: build123d parametric model + MASTER_PARAMETERS.json

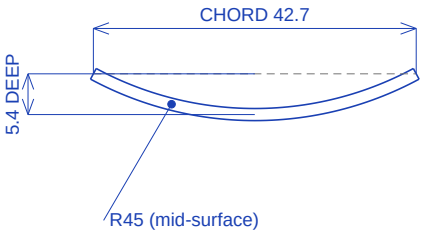
SHEET 10 - BLADE: FLAT PATTERN 1:1, CUP GAUGE 1:1, FORMING



Tabs CENTRED on the width; the blank is symmetric (no left/right version).

CUP GAUGE 1:1

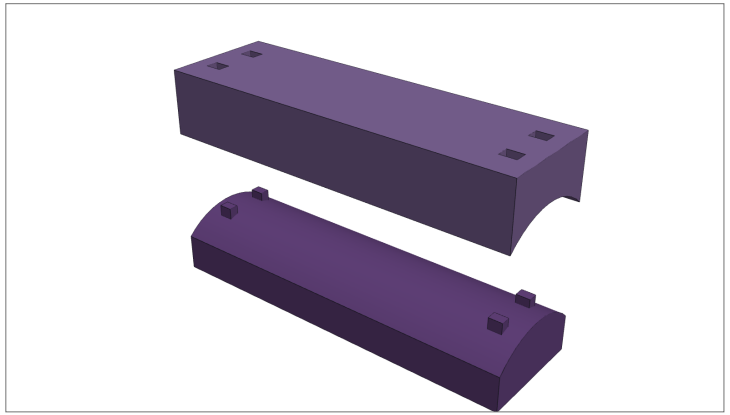
stand the formed blade on its end here



CUT (gate C: ONE blade first)
- 4 blanks per 11.75 x 3.75 in piece: rip 2 strips 44.5 wide, crosscut each at 139.7; kerf on the WASTE side.
- Notch tabs: saw/snips to 2 mm short of the shoulder, file R2 roots (no sharp inside corners).
- Break all edges; R2 outer corners; 1x45 tab tips.
- No holes in the blade.

FORM ACROSS THE WIDTH (printed cylinder die, vise)
- Die radius 33.3 = cup R45 / springback 1.35.
- Tabs sit between the 4 posts; shoulders butt them.
- Squeeze slowly until the die faces meet; release.
- Check on the cup gauge: 5.4 +/-0.7 mm deep.
- Too flat -> raise springback factor, reprint die.
- Blade must stay straight along its length.

ORIENTATION IN ROTOR
- All blades identical; concave side faces the same way round the ring (see top-view render).

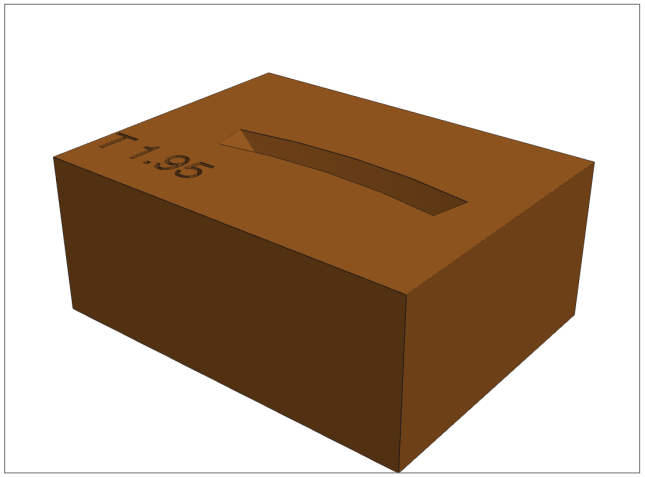


DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Blade

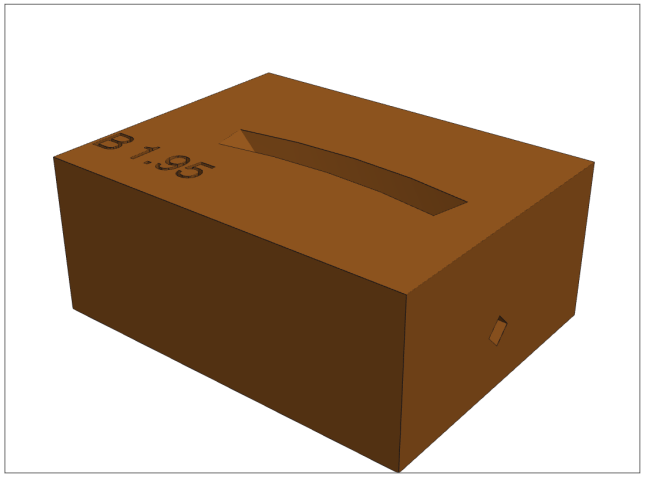
REV DW-V2.0 UNITS mm SCALE 1:1 (blade) 2026-10-24

SHEET 10/12 Source: build123d parametric model + MASTER_PARAMETERS.json

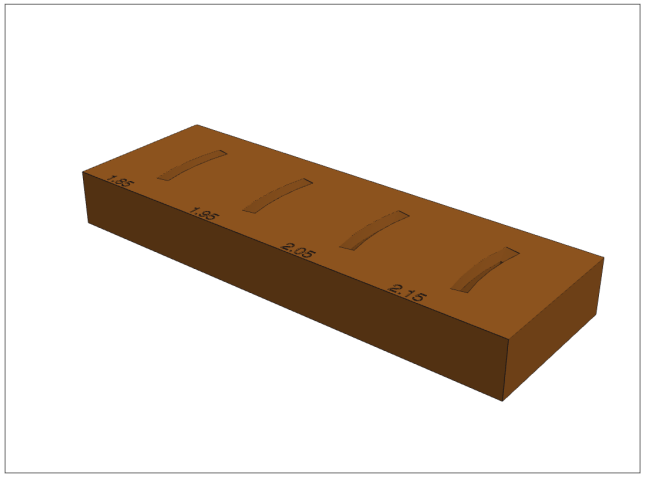
SHEET 11 - PRINT THESE FIRST: FIT COUPONS AND TOOLING



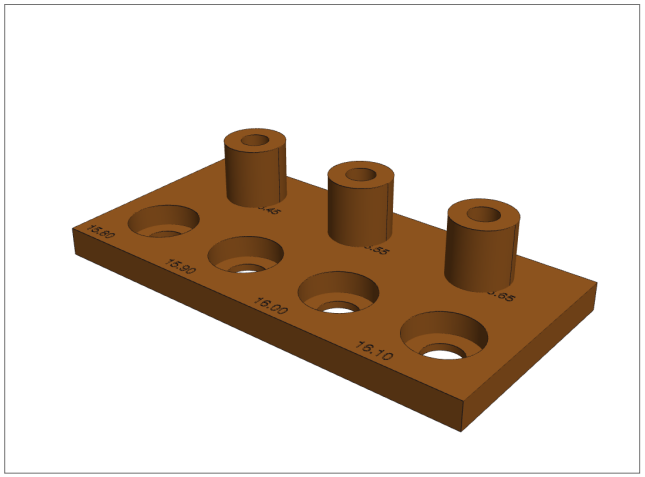
single_blade_socket_test_top (closed top, opens down)



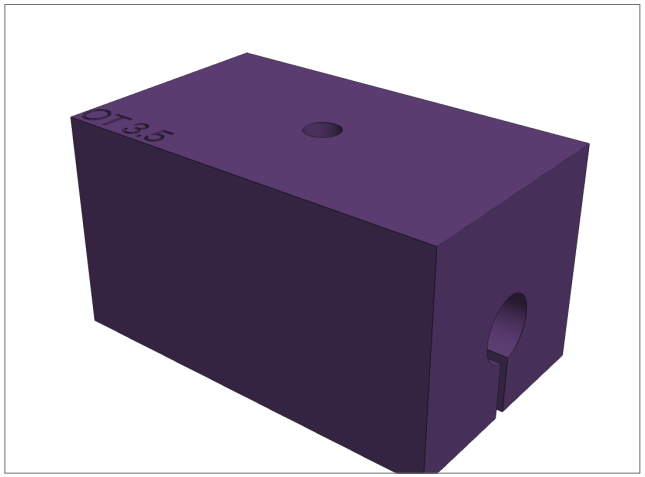
single_blade_socket_test_bottom (opens up, weep)



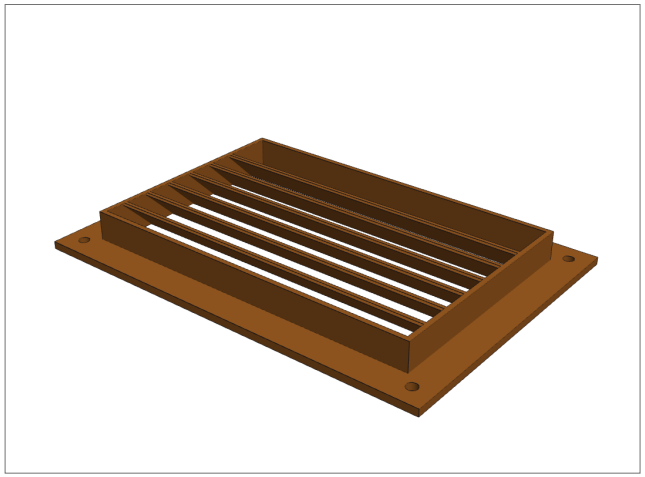
slot_clearance_coupon 1.85 / 1.95 / 2.05 / 2.15



bearing seats 15.80-16.10, shaft bores 6.45-6.65



shaft_dimple_drill_guide (3.5 mm spot)



inlet_vent_louver x2 (screened, 45 deg)

PASS CRITERIA (gate B): 1.6 mm aluminium tab slides in by hand, does not rock more than ~0.3 mm at the tip, no whitening/cracking or layer splitting when wiggled, cannot be pushed through any outer face. Choose the narrowest slot that passes, enter it in MASTER_PARAMETERS.json (pocket.width) and rebuild. Bearings: firm thumb-press, no rattle. Shaft bore: slides with light force, no wobble.

DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Coupons + tooling

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

SHEET 11/12 Source: build123d parametric model + MASTER_PARAMETERS.json

SHEET 12 - AUTOMATED CAD CHECKS, CALCULATIONS AND OPEN ITEMS

Calculation	Value	Comment
Blade mass (1.6 Al)	25.0 g	set of 20: 500 g
Centrifugal load @1000 rpm	31.0 N/blade	CG radius 113 mm
Per tab	15.5 N	bearing stress 1.4 MPa on 1.6x7 tab
Proof pull target	50 N per tab	= 3.2x the 1000 rpm load; keep
Pocket end-wall shear-out	> 1 kN est.	4 mm wall x 8.5 deep x 2 shear planes x ~15 MPa
Hub hoop stress @1000 rpm	< 0.1 MPa	negligible
Thermal (Al vs SS shaft, 60 K)	0.04 mm	cap separation change, << 7 mm engagement
Wind uplift on upper hub (25 m/s)	~5-10 N	resisted by set screw in dimple (positive)

OPEN ITEMS / NEEDS PHYSICAL VALIDATION

- Roof pitch is unknown: CAD built at roof_pitch_deg=0.0. Adapters for 0, 2/12, 3/12, 4/12 and 5/12 are exported; measure and rebuild.
- Roof deck thickness assumed 15 mm (only affects tower clearance below the deck).
- Pocket width 1.95 mm is a placeholder until the 1.85/1.95/2.05/2.15 coupon is tested with your actual aluminium.
- Bearing seat bore 15.95 mm and hub shaft bore 6.55 mm are placeholders until the bearing/shaft fit coupon is tested.
- Bearing bore not verified: confirm the R4-2RS markings and that the chosen shaft slips through both inner races without force.
- Forming-die springback factor 1.35 is a first guess; the alloy/temper of the 1.6 mm sheet is unknown.
- Cap separation 125.70 mm = blank body length; cut blanks to length within +/-0.2 mm. The tie rod pulls the upper hub onto the shoulders, so small length differences are taken up as light compression.
- Cup radius 45 mm: the 1:1 cup gauge checks it; 20 blades need 5 pieces of 11.75 x 3.75 in stock (4 blanks per piece).
- Shaft must be HOLLOW (your 1/4 in tube, ID 4.54) so the M4 tie rod passes; verify the tube slips into both bearings.
- Airflow, rpm and rain ingress are not predicted by this CAD; they must be measured (spec gates E and F).

MESH VERIFICATION (trimesh)

STL	watertight	bodies	dV %
upper_hub_closed_top.stl	yes	1	-0.046
lower_hub_closed_bottom.stl	yes	1	-0.074
thrust_washer_top.stl	yes	1	-0.06
thrust_washer_bottom.stl	yes	1	-0.06
bearing_support_stationary.stl	yes	1	-0.065
rain_baffle_if_needed.stl	yes	1	-0.047
blade_forming_die_lower.stl	yes	1	-0.041
blade_forming_die_upper.stl	yes	1	0.048
single_blade_socket_test_top.stl	yes	1	0.002
single_blade_socket_test_bottom.st	yes	1	0.004
slot_clearance_coupon_1p85_1p95_2p	yes	1	0.001
bearing_and_shaft_fit_coupon.stl	yes	1	0.002
shaft_dimple_drill_guide.stl	yes	1	0.004
inlet_vent_louver.stl	yes	1	0.0
lower_hub_SOLID_DISC_ALTERNATE_not	yes	1	-0.064
roof_flash_adapter_pitch_0in12.stl	yes	1	-0.06
roof_flash_adapter_pitch_2in12.stl	yes	1	-0.037
roof_flash_adapter_pitch_3in12.stl	yes	1	-0.036
roof_flash_adapter_pitch_4in12.stl	yes	1	-0.027
roof_flash_adapter_pitch_5in12.stl	yes	1	-0.027

DOGHOUSE WHIRLYBIRD - 20 CUPPED-BLADE ROTOR
Checks + calcs

REV DW-V2.0 UNITS mm SCALE NTS 2026-10-24

 Source: build123d parametric model + MASTER_PARAMETERS.json